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Impact of the KWL reading strategy on mathematical problem-solving achievement of primary school 4th graders

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ABSTRACT

In this study, it was aimed to investigate possible impact of using the KWL reading strategy in teaching the topic of “fractions and operations with fractions” to the 4th graders on students’ problem-solving achievement. A quasi-experimental design with pretest-posttest control group (CG) was employed in the study. The study group consisted of 35 4th graders studying at a primary school in Bartın, Turkey. In the research, an experimental group (EG) and a control group (CG) were used. There were 18 students in the experimental group (EG) and 17 students in the control group (CG). The test of equivalency (TE), Problem-Solving Achievement Test (PSAT) to gauge effect of instruction with the KWL strategy and reading texts were employed as data collection tools. The Wilcoxon Signed Rank Test and Mann Whitney U Test were employed in the research. It was found as a result of the research that the KWL strategy fostered the students’ problem-solving achievement. However, there were no significant differences between the EG and CG in terms of the scores received in the step of implementing the plan.

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problem-solving; problem-
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Introduction

Our age necessitates the development of individuals’ mental skills such as understanding, sorting, classifying, questioning, associating, predicting, analyzing, synthesizing and evaluating. Besides, raising individuals who are critical and creative thinkers, who are entrepreneurs, who can express themselves, who can communicate and who have advanced problem-solving skills is among educational policies of every country. In recent years, various approaches have been proposed for curriculums in Turkey, and improvements have been made from time to time. In this context (Ministry of National Education (MoNE), MoNE, MoNE, 2013; MoNE, 2018), the skills required to be acquired in mathematics curriculum include communication, reasoning, association, using information and communication technologies and problem-solving. As at all levels of education, it is aimed to develop problem-solving skills in primary education (Baykul, 2014).

Aim of this study was to examine impact of the KWL reading strategy on primary school 4th graders’ problem-solving achievement.

Theoretical framework

Constructivism, which is a philosophical approach that explains how individuals learn and understand, asserts the idea that knowledge is not explored but interpreted, is not revealed but created, that is, knowledge is constructed by individuals. Knowledge is shaped as a result of one’s own

experiences, observations, interpretations and logical thinking (Özden, 2009). Piaget, one of the pioneers of the constructivist approach, explained knowledge construction as a result of the active interaction of the individual with his environment through successive processes called accommodation and assimilation (Baki, 2014). According to the constructivist approach, learning is defined as “the process of constructing meaning by connecting new knowledge with old knowledge and experience”. The learner constructs his/her own conceptualization and solutions of problems. This approach acknowledges that students learn new knowledge by adapting the information they have already learned to their subjective situations (Özden, 2009). In this context, constructivism is based on Ausubel’s “Meaningful Learning Theory”. According to Ausubel, for a meaningful learning, the individual should have the correct prior knowledge about the subject to be learned and be willing and determined to learn, as well as associating the new knowledge and the prior knowledge (Ausubel, 2000, as cited by Meydan, 2018).

According to NCTM (2000), pupils should learn mathematics by understanding it and actively construct new knowledge by making use of their prior knowledge and experiences. Principles and Standards for School Mathematics’s (2000) learning principle is based on two basic thoughts the first of which is learning mathematics by understanding it. Today’s mathematics requires not only computational skills, but also the ability to reason and think in order to solve new problems, and to learn new ideas that students will encounter in the future. The second is that

students can learn mathematics by understanding. Learning is developed in classrooms so that students can evaluate their own and others' ideas, and they are encouraged to generate and test mathematical assumptions and to develop reasoning skills (as cited by Van De Walle et al., 2013).

Mathematics is commonly regarded as a difficult subject to be understood by students. To learn mathematics, firstly it should be understood. For this, it is significant for a student to understand what he/she reads and make out it. There are some basic principles that should be taken into account in addition to reading comprehension for understanding mathematics. One of them is concept teaching. It is impossible for students to learn mathematics without understanding concepts. Understanding a concept is associating it with existing concepts. The more a concept is associated with the concepts it is related to, the better it is understood. Learning mathematics requires establishing a relationship between conceptual knowledge and operational knowledge (Baykul, 2014). The development of mathematical thinking depends on the conceptualization of mathematical knowledge and its use when necessary. Good knowledge of mathematical concepts facilitates to find unique solutions to new problems that have not been encountered before (Özyıldırım Gümüş & Umay, 2018). Mathematical models, various proofing methods, and examples from mathematics itself and from daily life can be used to understand mathematics. Examples used in accordance with various teaching objectives enrich and strengthen learning (Kula & Ören Vural, 2019). In addition, the concept of proof can be used as a tool in understanding the concepts and the relationship between the concepts (Knuth, 2002). On the other hand, in the process of problem-solving, the student can get the opportunity to see the implementation of mathematical concepts and connect the concepts with the method of operation, algorithm and equation to be used for the solution (Schoen & Charles, 2003, as cited by Baki, 2018). In the process of problem-solving, the individual has to use his existing knowledge actively. In particular, it is necessary to use conceptual and operational knowledge in association. For students, learning mathematics is possible by connecting the prior knowledge with new knowledge. When this does not occur, it may not be possible to understand and learn mathematics. (Hiebert & Carpenter, 1992, as cited by Özyıldırım Gümüş & Umay, 2018).

According to Charles (1985), for solving a problem, the student goes through cognitive processes such as understanding the problem sentence, selecting the necessary data for solution, generating a solution to the problem, and deciding whether the solution is logical or not. In this process, the student gathers concepts and processes and applies them to solve the problem (Bernardo, 1999, as cited by Baki, 2018). In the process of problem-solving, the student becomes aware of his/her knowledge, investigates for solutions to the problem by using his/her prior knowledge, makes predictions and evaluates his/her solutions. In this situation, the student uses his/her existing knowledge while investigating for solutions to the problem, explains, analyzes, organizes and makes sense of all information (Baki, 2018).

The development of mathematical thinking depends on the development of mathematical skills. Problem-solving is an important mathematical skill, too. Understanding the problem, generating a strategy for solution, implementing the strategy, evaluating solutions and writing a similar problem are the steps included in this process. Understanding the problem is the most important step of problem-solving process. If a problem cannot be identified, a solution cannot be suggested and the problem cannot be solved. In the step of understanding the problem, the most important critical behaviors are to write down the things given and asked in the problem, to explain the problem with the students' own expressions and to draw a figure or diagram explaining the problem (Baykul, 2016). It is highly critical for students to understand what they read and to explain the problem in their own words at this step.

Not understanding the problem properly causes students to have problems. Stating the problem in the students' own words and reading it as required are two important sub-steps in understanding the problem (Baykul, 2016). According to Ogle (1986), since the KWL reading strategy includes activities related to determination of the students' prior knowledge about the text to be read, what they know and what they want to know, the student will be able to connect the old knowledge with the new one. According to Stephens and Brown (2000), for a student to connect the knowledge before the reading process with the information obtained after reading enables him/her to organize, integrate and summarize (as cited by Ulu et al., 2016). For this reason, based on the expressions of Ogle (1986) and Stephens and Brown (2000), the idea that the KWL reading strategy would be effective in problem-solving was determined as the aim of the study.

A student's solving a problem depends on his/her understanding it well. For providing this, he/she needs to fully comprehend what he/she reads. Reading comprehension means that individuals reach a new learning as a result of associating new information they encounter in the texts with their prior knowledge (Yılmaz, 2019a). Reading comprehension process occurs through conscious thinking, which helps the reader relate thinking processes, content of the text, prior knowledge, expectations and reading purposes (Block, 2004, as cited by Erdem, 2016). Reading comprehension can be regarded as a process of creating a new related situation by linking the words with schemas in mind. In this process, mental activities such as thinking on a text or problem, searching for the reasons, drawing conclusion, interpreting, analyzing, synthesizing and evaluating are influential (Güneş, 2000). KWL, one of the reading strategies, contains activities to develop reading comprehension skill. The KWL table used for this strategy is a reflective table indicating what they know, what they want to know and what they learned. Using this strategy helps the students think on a topic before reading it and the teacher gauge what his/her students know after reading (Nicely, 2015). The KWL strategy is quite effective in developing reading proficiency. A great number of studies have revealed that this strategy is useful for developing students' reading skills and raising

Table 1. The KWL chart.

KWL Chart		
K-What I Know?	W-What I Want?	L-What I Learned?

(Source: Ogle, 1986).

their achievement (e.g.: Akily, 2010; Alsalhi, 2020; Hana & Faridi, 2015; Riswanto et al., 2014; Lou et al., 2016).

The KWL (What i know, what i want to learn, what i learned) strategy

KWL is one of the strategies used for improving reading comprehension skill. This strategy was developed by Ogle (1986). This strategy was named based on the initials of “What I know, What I want to learn, What I learned”. This strategy provides students with the opportunity to link with their background knowledge before reading and the knowledge gained during reading. Table 1 shows the chart on which the students listed their knowledge during implementation of the KWL strategy.

A three-column table is prepared before starting reading in this strategy and the students brainstorm ideas to reveal all they know about the topic or problem. This information is written in column K, which is “What I know”. Then, the students prepare and save questions related to what they want to find out about the topic or problem in column W, which is “What I want to learn”. During or after reading, the students answer the questions in column W and save their answers in column L, which is “What I learned” (Epçaçan, 2009; Yılmaz, 2019b). The KWL strategy includes activities to reveal what background knowledge the students have about the text to be read prior to reading, what they know and what they want to know. This strategy consists of three steps as follows:

1. **Step K – What I know:** In this step, the students’ background knowledge and views on the new topic to be read are concretely revealed (Hill et al., 1998). The students are led to brainstorm (they are asked questions about the topic) to reveal what they know about the topic for reading. In this way, the students are provided with textual connections before obtaining detailed information about the text they will read. After all the students know about the topic are revealed, this information obtained is written in column “K” by the students.
2. **Step W – What I want to learn:** In this step, the students set personal goals for their own learning (Carr & Ogle, 1987). They create lists of questions regarding what they want to learn at the end of the text to be read. They record the questions created in column “W”. Before the students start reading, determining the purpose of reading, knowing what the text to be read will provide them, considering why they are reading, can attract their attention to the text (Yılmaz, 2008).
3. **Step L – What I learned:** In this step, the students read the text. They answer the questions written in column “W” during or after reading. In this way, the students

construct the meaning by associating the information they have newly learned with their prior knowledge. After having finished answering the questions, they record their answers in column “L”.

The KWL strategy has been found effective in developing reading skills by a great number of studies (e.g.: Eker, 2015; Peköz, 2018; Rahayuningsih & Wulandari, 2014; Tokgöz, 2019; Özyılmaz & Alcı, 2011; Zouhor et al., 2016). Riswanto et al. (2014) stated that the students could not read because of their deficiencies in reading comprehension and were not able to relate the text they read with their prior knowledge. Active readers question for applying their schemas during pre-reading activities, relating them with the new concept, understanding the topics during and after reading and making their own meanings (Alyousef, 2005; Puti, 2014; as cited by Wijaya & Puti, 2015). Moreillon (2007) said that “reading is making meaning from print and visual information”, and essence of reading is to understand the message of a text or an author accurately, thus reading comprehension requires higher-order thinking (as cited by Wijaya & Puti, 2015).

Barton and Heidema (2002) answered the question “why should students learn to read mathematical texts?” as follows by citing from Martinez and Martinez (2001) and NCTM (2000). Children discuss what happens when they read and write mathematics. Therefore, mathematical language and mathematical reading should be learned in order to focus on the problems, work on them, convey ideas consistently and explicitly, arrange arguments, develop interpreting flexibility and understand own problem-solving and thinking processes. Mathematical texts contain more concepts than all other types of texts. As most of these concepts are abstract, it becomes more difficult for students to understand and visualize them. Students generally look for a page of text including examples, graphs and problems to solve by skipping statements of important information (Thomas, 1988; as cited by Barton & Heidema, 2002). On the other hand, mathematical texts are often very short and compact, and each sentence includes a lot of concepts and information. The fact that the sentences have certain meanings and connect to other sentences sensibly may cause students wanting to read these texts swiftly not to notice important details, explanations and underlying logic (Barton & Heidema, 2002). Cognitively, reading is more than reading a text aloud with correct pronunciation. Active reading contains thinking on the things read. Reading comprehension requires a reader’s synthesizing information in a text, interpreting, evaluating and using it selectively (Wijaya & Puti, 2015). The mathematical literacy processes are necessary both for problem-solving and developing concepts. These skills are necessary for solving mathematical problems as well (Barton & Heidema, 2002). This is because of the fact that determining, understanding and questioning the problem is primarily needed to solve a problem. The questioning can be started with determining if there is incomplete or extra information. Questioning helps to distinguish the information unnecessary for solving the problem, to determine the information necessary for solving it and to create a

solution algorithm accordingly. Via questioning can individuals make a plan for solving the problem and solve the problem if they have know-how – that is topic knowledge. It can also contribute to decide whether solution of a problem is sensible and correct. It was expressed that the KWL strategy encourages pupils to be mentally active during the learning process, facilitates learning and remembering concepts and topics, develops their skills in organizing their prior knowledge and in questioning (Alsalihi, 2020). Therefore, it can be considered as a metacognitive strategy (Gammill, 2006). the KWL is a strategy used by teachers to help students activate their schemas prior to reading, improve comprehension during reading and organize their thoughts after reading (Piper, 1992, as cited by Alsalihi, 2020).

Shelly et al. (1997) stated that a mathematics teacher used the KWL strategy successfully to teach mathematical operations; Siribunnam and Tayraukham (2009) mentioned that the KWL strategy generally promote pupils to use thinking skills and concentrate on analytical reading; Ibrahim (2012) indicated that students can relate old and new knowledge by using the KWL strategy actively. According to Szabo (2006), using the KWL strategy has several advantages which are increasing the students' interest in topics and motivating them to understand by activating their prior knowledge related to the topic through brainstorming. Additionally, students can restrict topics, increase their reading comprehension, learn by their own and expand their ideas and perceptions regarding the topic via this strategy (Szabo, 2006). Mathematics is related to problem-solving, and reading comprehension is an important component of word problems (Heidema, 2009). It is known that reading comprehension develops thinking in all areas. Since reading is an active process in which important ideas are detected, compared, evaluated and related, it is closely related to mathematics (Erdem, 2016). Individuals that do not have difficulty in reading comprehension are generally able to learn to express their opinions, to gain a different perspective, to write, to do mathematical operations and to make connections for problem-solving (Riedel, Tomaszewski, and Weaver, 2003, as cited by Erdem, 2016).

Studies have revealed that reading comprehension skills are crucial for students to solve mathematical problems (Göktaş, 2010; Lerkkanen et al., 2005; Pape, 2004; Passolunghi & Pazzaglia, 2005; Vilenius-Tuohimaa et al., 2008). Nicely (2015) indicated that instruction merging problem-solving strategies and KWL strategies increased students' success in solving mathematical problems. Passolunghi and Pazzaglia (2005) emphasized that it is necessary to have a good level of reading comprehension to be a good problem-solver. A strong correlation was found between the skill of mathematical problem-solving and of reading comprehension by Vilenius-Tuohimaa et al. (2008), between reading comprehension skill and critical thinking skill by Aloqaili (2012) and between reading comprehension and mathematics success by Göktaş (2010). On the other hand, Knopik and DeFries (1999) concluded that reading

and mathematics interact and deficiencies are revealed together.

While Akyüz (2004) highlighted that the KWL strategy increased ninth graders' achievement, Taşlıdere and Eryılmaz (2012) emphasized that intervention merging reading strategy and conceptual physics approach raised student achievement significantly. Zouhor et al. (2016) found the KWL strategy to be efficient in increasing the sixth graders' metacognition and physics success. Alsalihi (2020) revealed that the KWL strategy contributed positively to the 4th graders' success in science courses. Phromphithak (2015) investigated impact of the KWL strategy on 10th graders' mathematics achievement. As a result of the research, she found that the KWL strategy was effective in pupils' understanding mathematical concepts, improving their abilities to solve mathematical problems and promoting their mathematics achievement.

There have been studies investigating the effect of reading comprehension skill on problem-solving skill in different dimensions and from different perspectives, and their results were positive. Ulu et al. (2016) found that instruction of reading comprehension and problem-solving strategies implemented on 5th graders fostered the students' problem-solving achievement, reading comprehension strategies could explain just 7,86% of the problem-solving achievement, and the pretest scores could explain 87,80% of the change. Thus, Ulu et al. (2016) concluded that students' readiness levels were more influential in predicting problem-solving achievement than their reading comprehension strategies. Vilenius-Tuohimaa et al. (2008) revealed in their study that fluent reading skill affected comprehension skill directly and problem-solving skill through comprehension skill. They also stated that there is a high correlation between comprehension skill and problem-solving skill.

It was stated that good problem solvers have high comprehension levels (Prakitipong & Nakamura, 2006; Tertemiz, 1994), and their reading comprehension skills increase their effect on problem-solving achievement over the years (Grimm, 2008). According to Fuentes (1998) good problem solvers use mental networks to create meaning, so they can relate events in the problem text and follow the flow of events. Grimm (2008) revealed in his study that reading comprehension skill gained in 3rd grade is a predictor of the development of problem-solving achievement until the end of 8th grade. In the studies examining achievement of students who make word-based and action-based meanings (Hegarty et al., 1995; Moreno and Mayer, 1999; Panasuk and Beyranvand, 2010), it was seen that the latter ones were more successful. Because in action-based comprehension, the event fiction in the problems is analyzed rather than the words, the relationship between the events is analyzed and the entire problem text is focused (as cited by Ulu et al., 2016). In his study, Hite (2009) tried to eliminate the 5th graders' deficiencies related to their problem-solving skills by giving them reading comprehension training and stated that achievement of the students with reading comprehension difficulties increased at the end of the study. Due to the similarities between understanding stories and

understanding word problems, Fuentes (1998) stated that reading comprehension strategies can also be used to understand the problem stories. Özdemir and Sertsöz (2006) stated that providing middle school 6th graders with reading comprehension behavior had a positive effect on their mathematics achievement.

Ilgın and Arslan (2012) examined the effects of problem-solving instruction through text-based activities on problem-solving achievement levels of primary school second graders. As a result of their studies, they concluded that problem-solving instruction with texts was useful for helping the students gain the steps of problem-solving, which were “determining possible solutions in the light of the collected data, practicing the determined solution, evaluating solution types, assessing the problem-solving method employed”.

Ulu (2017) investigated the impact of reading comprehension and problem-solving strategies of primary school 4th graders on classifying the students by high and low problem-solving achievement. As a result of the study, it was found that fluent reading skills had no effect on classification of students by high and low problem-solving achievement. In addition, it was stated that the students with high problem-solving achievement had high levels of comprehension, yet there was no significant difference between the students with high and low problem-solving achievement in terms of having basic skills of mathematics.

Solving mathematical word problems requires using both mental representation skill and reading comprehension skill. Boonen, Koning, Jolles, and Van der Schoot (2016) stated that reading comprehension training and a realistic mathematics education approach had positive effects on middle school 6th graders’ ability to solve mathematical word problems and that developing reading skills for solving mathematical word problems should be attached great importance. In a study conducted by Nicolas and Emata (2018) to examine the development of problem-solving skills of middle school 7th graders by employing the comprehension strategy with an integrative approach in mathematics teaching, a significant difference was found in favor of the experimental group.

Öztürk et al. (2020) found in their study, in which they investigated the impact of reading comprehension skills on mathematics self-efficacy perception, mathematics attitude and the ability to solve non-routine mathematical problems, that there was a positive and significant correlation between reading comprehension skills and other variables. There is a positive and significant correlation between reading comprehension skill and problem-solving skill.

There have been numerous studies, some of which were described in detail above, showing that there is a positive and significant correlation between students’ reading comprehension skills and problem-solving skills (Boonen et al., 2016; Çavuşoğlu, 2010; Kılıç, 2009; Pape, 2004; Ulu et al., 2016; Uzun, 2010). It can be understood from these studies that reading comprehension skills need to be developed in early childhood (Silver & Cai, 1996; Helwig et al., 1999, as cited by Boonen et al., 2016).

Problem-solving which is one of the methods that helps bringing information to life, activates the students in daily life and learning-teaching settings and encourages the students to learn to learn (Özcan, 2007) is defined as the way of reaching a solution when the solution is unknown (Polya, 1962). Thus, mathematics, itself, is considered to be a problem-solving activity. Therefore, problem-solving has been included in curriculums as one of the basic approaches in mathematics teaching (MoNE, 2013, 2018; National Council of Teachers of Mathematics (NCTM), 1989; Silver, 1994). It is necessary to apply reasoning in the process of problem-solving (Schoenfeld, 1985). Reasoning is the process of thinking by considering all factors and obtaining a reasonable result. Individuals who are able to reason about a topic have sufficient knowledge in that topic, can examine new situation encountered thoroughly, explore, guess, infer, reach to some results by justifying thoughts, explain these results and offer solutions (Umay, 2003). Hence, a problem cannot be solved unless it is understood well, and mathematical ideas cannot be defended without showing reasons. Although most people believe that there is only one way to solve a mathematical problem (NCTM, 2000), each individual has his own reasoning styles and ways of solution (Umay, 2003). In this context, students’ knowledge can be questioned and interpreted as they compulsorily reason during the process of problem-solving. In this process, the students are expected to apply cognitive processes such as understanding a problem, collecting the necessary data for the solution, solving the problem and deciding whether the solution is logical or not (Charles, 1985). The students have to gather the concepts and operations during problem-solving and apply them to solve the problem (Bernardo, 1999). Polya (1973) explained problem-solving process in steps which are understanding the problem, devising a suitable plan for solving the problem, changing the plan if necessary, checking the methods, evaluating the data obtained during problem-solving and interpreting significance of the solution. On the other hand, Gonzales (1998) suggested that a final step, problem posing or problem writing, was necessary to be added to Polya’s problem-solving strategy. In this context, five-step grading scale received from Baki (2014) was employed in the current study in order to evaluate each step separately.

From the explanations made in the theoretical framework and the results of the studies, some inferences can be made in the context of the KWL reading strategy and problem-solving. The first of these inferences is the way of implementing the KWL reading strategy in classroom setting. A KWL table is created before starting a new topic, and this table can be filled in as a class or individually. Meanwhile, the teacher should remind the students to use mathematical concepts and terms. While filling in the table, the students should be asked questions that would help them explain what they have learned, how they have solved problems, and their reasoning strategies. Secondly, the KWL reading strategy helps students to activate their prior knowledge, organize their knowledge and summarize the topic in the learning process. In this process, the students share various ideas by

brainstorming, while the teacher facilitates learning by asking them some questions. Moreover, the students organize, analyze and interpret their knowledge by writing down what they know and learn. Because of these features of the KWL strategy, it can be used as an effective teaching instrument in the problem-solving process. This stems from the fact that problem-solving also requires organizing knowledge, developing various solution strategies to the problems and finding solutions. The problem-solving process requires understanding the problem, choosing appropriate strategy for the solution, implementing the chosen strategy, evaluating the solution, and using the solution in similar situations. The problem-solving process requires the students to use a variety of reasoning and problem-solving strategies. The studies cited in this study have shown that development of reading strategies generally improves students' understanding of mathematical concepts and their ability to solve mathematical problems. In this study, impact of the KWL reading strategy on solving mathematical problems in each of the problem-solving steps in the problem-solving process was investigated.

Teaching of fractions has an important place in mathematics curriculums (MoNE, 2013, 2018). Strong conceptual knowledge about the concept of fraction is important in the development of the concept of number (Lazić et al., 2017). Due to their conceptual richness and complexity, fractions are among one of the most difficult topics to learn in mathematics (Alacaci, 2010; Işık, 2011; Işıksal, 2006). Understanding the concepts of fractions well in primary school and gaining the ability to do operations with fractions will make the students' daily life easier and will provide them a basis for learning other topics of mathematics such as ratio, operations, rational numbers, decimal numbers and algebra (Alacaci, 2010). However, the studies in the literature have indicated that students from primary school to university have several misconceptions about fractions and experience various difficulties (Alacaci, 2010; Biber et al., 2013; Kocaoğlu & Yenilmez, 2015; Misquitta, 2011; Okur & Çakmak Gürel, 2016; Pesen, 2007, 2008; Stafylidou & Vosniadou, 2004; Taşkın & Yıldız, 2011; Vamvakoussi & Vosniadou, 2004). It was stated that a significant part of misconceptions about fractions stems from the generalization of the information valid in integers to the topic of fractions (Alacaci, 2010). In addition, teaching students numerical representation and operations about fractions earlier without teaching them the important conceptual elements of fractions (Bezuk and Bieck, 1993, as cited by Alacaci, 2010) causes them to have difficulties in learning the topic. Therefore, the topic of fractions which is really important in mathematics curriculums due to its being the basis for teaching other topics of mathematics and to its importance in daily life was discussed in the current study.

Mathematics is about problem-solving, and the abstract structure of mathematics mostly makes it difficult to be understood by the students. Reading comprehension is a higher-order skill to be gained for solving mathematical problems. Reading comprehension necessitates thinking - that is reasoning and questioning. Problem-solving; on the

other hand, requires reasoning. Reading comprehension is the first and foremost step to solve a mathematical problem. In daily life or in order to solve a mathematical problem, firstly the problem ought to be revealed, in other words, it is needed to be understood. Understanding the problem means that the problem is properly read and expressed. Establishing the link between what is given and what is asked in a problem is necessary for solving a mathematical problem.

KWL, one of the reading comprehension strategies, includes activities to be used for developing reading comprehension skill. There have been several studies conducted on reading comprehension skill and mathematics achievement and problem-solving skill. However, no studies examining effects of the KWL strategy on 4th graders' problem-solving achievement and using problem-solving steps separately and thoroughly for solving problems about fractions have been found in the literature. As it was expected that the KWL reading strategy would be effective in understanding the problem and developing an appropriate problem-solving strategy because it establishes a relationship between old and new information, it was used in the present study. From this aspect, impact of the KWL reading strategy on primary school 4th graders' mathematical problem-solving achievement was investigated. It is believed that the results of this study would contribute to the literature.

In this study, impact of the KWL reading strategy on mathematical problem-solving was examined by considering each of the problem-solving steps thoroughly. One of the two reasons for this was the fact that knowing at what level the students put forward their problem-solving steps, the processes and thoughts they have carried out at each step of the problem-solving process and the steps they have difficulty provides the teachers with the opportunity to know their students more closely (Baki, 2018). In each of the problem-solving steps, there are various procedures to be followed by the students. It is believed that knowing in which of the problem-solving steps the student has difficulty is important in terms of the precautions to be taken in the learning process. The second reason was that problem-solving covers the whole process from understanding the problem to checking the accuracy of the result obtained. In this study, impact of the KWL strategy on students' achievement in each step of problem-solving and their overall problem-solving achievement was investigated.

Problem statement of the current research was to reveal "impact of the KWL reading strategy on mathematical problem-solving achievement of primary school 4th graders". Aim of the study was to investigate impact of the KWL reading strategy on primary school 4th graders' problem-solving achievement.

Aim of the study

This research aimed to examine impact of using the KWL reading strategy to teach the primary school 4th graders the learning area of "numbers and operations", the sub-learning area of "fractions" and the topic of "fractions and operations

Table 2. Model of the research.

Groups	Pretest		Implementation	Posttest	
	Test of equivalency	Problem-Solving Achievement Test		Problem-Solving Achievement Test	
EG	✓	✓	Instruction with the KWL Reading Strategy		✓
CG	✓	✓	Instruction with Current Instructional Program		✓

with fractions” on the problem-solving achievement of students. Accordingly, sub-problems of the study were determined as follows.

Sub-problems of the study

1. Are there any statistically significant differences between problem-solving achievement pretest scores of the EG students on whom the KWL strategy was employed and of the CG students on whom current instructional program was employed?
2. Are there any statistically significant differences between problem-solving achievement posttest scores of the EG students on whom the KWL strategy was employed and of the CG students on whom current instructional program was employed?
3. Are there any statistically significant differences between pretest and posttest problem-solving achievement scores of the EG students on whom the KWL strategy was employed?
4. Are there any statistically significant differences between pretest and posttest problem-solving achievement scores of the CG students on whom current instructional program was employed?

Methodology

Model of the research

A quasi-experimental design with pretest-posttest control group was employed in the study. This design provides great statistical potential to the researcher about testing the effect of intervention on dependent variable, and helps interpretation of findings obtained within the context of cause and effect (Büyüköztürk, 2011). The tests used in the experimental group (EG) and in the control group (CG) during the study were given in Table 2. Accordingly, in both of the groups, test of equivalency (TE) and Problem-Solving Achievement Test (PSAT) were employed as the pretest while Problem-Solving Achievement Test (PSAT) was used as the posttest. Implementation with the KWL reading strategy was used in the experimental group while implementation based on the current instructional program was applied in the control group. The empirical design of the research was given in Table 2.

Study group

The research was conducted in 2018–2019 academic year. The study group consisted of thirty five 4th grade students studying at a primary school in Bartın, Turkey. The EG

included 18 students, and the CG included 17 students. Both groups were taught in accordance with the learning outcomes of the topic of fractions included in 4th grade MoNE (2018). The ready-made groups were preferred to create the EG and CG. Equal groups were determined after the implementation of the test of equivalency, and one of them was assigned as the EG, while the other was assigned as the CG. Instruction was implemented using the KWL reading strategy in EG, while it was implemented using the activities in the current MoNE (2018) in the CG. The implementation lasted 20 lesson hours in each classroom.

Data collection tools

Test of equivalency (TE), Problem-Solving Achievement Test (PSAT) to gauge effect of instruction with the KWL strategy and reading texts prepared to be used in both groups were employed as data collection tools. The reading texts were chosen from primary school 4th grade Turkish coursebook.

Test of equivalency (TE)

A prior knowledge test prepared for primary school 4th graders by Güngör (2014) was used as test of equivalency in this research. This test was conducted in order to determine whether the groups were equal or not. The validity and reliability of the test had been provided by Güngör (2014), and the reliability coefficient had been found to be 0.71. The test was applied to both groups. The purpose of applying this test was to compare initial achievement levels of the EG and CG students. In the test of equivalency, there were questions about learning outcomes of addition, subtraction, multiplication, reading natural numbers, writing natural numbers, and greatness-smallness in natural numbers. The Test of equivalency was given in Appendix 1.

Problem-solving achievement test (PSAT)

A PSAT including 9 problems, which were analyzed for validity and reliability, about the topic of “fractions and operations with fractions” was prepared by the researchers benefitting from the literature (Baykul, 2016; Özçelik, 2018; Posamentier & Krulik, 2016; Yazgan & Arslan, 2017) in order to evaluate students’ problem-solving achievements. In the analysis of PSAT, each step of the problem-solving process that are understanding the problem, devising a plan, carrying out the plan, looking back and suggesting a problem was considered as a separate sub-problem. In order to ensure the content validity of the problems in the test before implementation of the PSAT, two content area experts, a



Figure 1. The empirical study process.

mathematics teacher and a classroom teacher working at a secondary school were asked to express their opinions about the test in terms of content, level and language. Then, PSAT was finalized, and a pilot study was conducted with 20 students to provide reliability of the research and to see availability of the data collection tool. Necessary editions were made as a result of the pilot study, and a lesson hour (40 minutes) was regarded enough to answer the questions. Thus, the problem-solving achievement test, as a data collection tool, consisted of 9 problems. The questions in PSAT were prepared in accordance with the learning outcomes of the topic of “fractions and operations with fractions” included in MoNE (MEB, 2018) which were “Students will be able to recognize simple, compound and integer fractions and to show them with models. They will be able to compare and to sort unit fractions. They will be able to determine a specified fraction of a quantity. They will be able to compare up to three fractions whose denominators are equal. They will be able to add and subtract fractions with equal denominators. They will be able to solve problems that require addition and subtraction with fractions”. Instruction was implemented using the KWL reading strategy in the EG, while it was implemented using the activities in the current curriculum in the CG. During the implementation of the tests, the students were given detailed explanations by the researchers and were asked to solve the given problems in a lesson hour by considering each step. In the present study, the students’ answers were evaluated by the progressive scoring scale, received from Baki (2014), in which each step used in problem-solving process was scored. This scale was preferred as the students’ behaviors of problem-solving and writing a similar problem were examined

separately. The progressive scale consists of five categories. Under these categories, there are four criteria (3, 2, 1, 0) describing the highest and the lowest performance of each criterion. Thus, the maximum score to be received from each step by a student would be 3, while the minimum score would be 0. Therefore, a student could receive 3 points at maximum and 0 point at minimum from a problem. The responses given to 9 problems by the students were scored by the progressive scoring scale, and a scoring in the range of 3-2-1-0 was made for each category in this scoring scale by the researchers. In this scoring, the scores received at each step by the students were evaluated separately. The Problem-Solving Achievement Test (PSAT) and the Progressive Scoring Scale (PSS) were given in Appendix 2 and Appendix 3 respectively.

Procedures

There were 18 students in the EG who were taught using the KWL strategy and 17 students in the CG who were taught in accordance with current instructional program. The data of 4 students in the CG were not included in the study since they had not attended the lessons regularly. The studies carried out during the empirical process were presented in Figure 1. Instruction was implemented according to the KWL strategy in the EG and to current Mathematics Curriculum in the CG by the primary school 4th grade class teacher.

Before starting the implementation, the 4th grade class teacher and the EG students were informed about the KWL strategy, and they were told about things to be done during the implementation. Instruction with the KWL strategy was

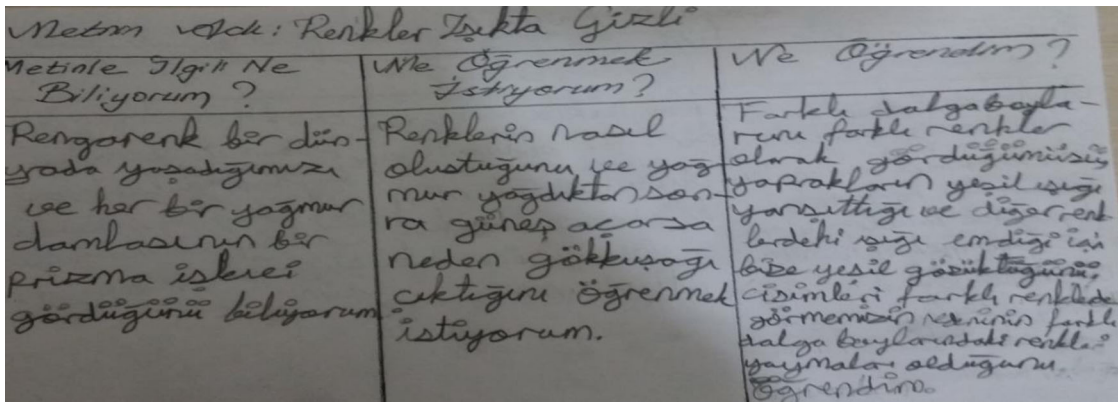


Figure 2. Study of a student in the EG about the reading text “Colors Hidden in the Light”.

carried out with 9 texts chosen from 4th grade Turkish coursebook. The reading texts were studied using the KWL strategy in the EG. These reading texts were as follows:

“A Memory of Mustafa Kemal, The Snake and Man, A memory of Atatürk, The World is Round, Virtual Illness, The Source in Nature, The Seed Looking for Home, Colors Hidden in the Light, The Great Legend”.

The topic of fractions was taught to the EG by the same teacher in the Math lessons using the KWL strategy. The implementation lasted for 20 lesson hours. The study of the two students in the EG about the reading texts “Colors Hidden in the Light” and “The World Is Round” were given in Figure 2 and Figure 3, respectively.

In the EG and CG, 13 mathematical activities about the topic of fractions and operations with fractions in primary school mathematics course book were implemented. However, the mathematical activities in the EG were carried out by using the KWL strategy. The mathematical activities performed by 2 students in the EG using the KWL reading strategy were given in Figure 4. The activities were carried out in mathematics and Turkish lessons synchronously. PSAT was applied to the EG and CG students as pretest and posttest before and after the implementation.

The KWL table is a reflective chart showing what students know, what they want to learn and what they learned (Ogle, 1986). The teacher firstly asked the students to create their own KWL charts after giving them brief information about the KWL. At the beginning of the implementation, the students were asked questions about the topic in order to reveal their prior knowledge via brainstorming. The students were required to remember their background knowledge and take notes by doing brainstorming. They wrote this knowledge on ‘K’ column of the KWL chart. After they became familiar to the problem, the problem statement was read and definition of the problem was made. In this step, the given and the asked things in the problem were identified, and the students were asked to draw a figure or schema related to the problem. They moved to the step of understanding and interpreting the knowledge with this step. They wrote what they wanted to learn or thought they knew on ‘W’ column of the KWL chart. In these two steps, they started to link the knowledge they knew or thought they knew to the knowledge they wanted to know. After reading the problem and entering into the process of

interpretation, the students wrote what they learned from the problem on ‘L’ column, which is ‘what I learned’. By this way, the students, who organized and arranged their knowledge moved to the second step of problem-solving – devising a plan. Devising a plan is creating a solution strategy for the problem by organizing the data and identifying all information included in the problem. This step contains remembering the current problem, similar problems and solutions. In this case, it means that the student has established a link between his/her prior knowledge and new knowledge. If the student has strategy knowledge to carry out the plan he/she has devised, he/she can find a solution by carrying out the plan. In the step of carrying out the plan, the students were asked to question the given statement, perform foreseen solutions regarding the problem, check if their solutions worked or not - or if they were correct or not, express their reasons for proving that their solutions were correct. In the final step, they were required to write a new problem similar to the given problem. The implementation was performed sometimes individually and sometimes by creating a discussion environment.

An experimental group student’s solution found by using the KWL strategy was given in Figure 5. In order to understand the implementation process better, sample explanations that the student made about use of the KWL strategy in solving the problem and interpretations were included in Table 3.

In Table 3, it was seen that the student wrote what he knew about the topic on ‘K’ column and recalled prior knowledge – the basis for solving the problem – from mental schema. In ‘W’ column, the questions that would help problem-solving and that are needed to be asked in the steps of mathematical problem-solving were asked. The student had been asked to write interrogative expressions in ‘W’ column. In ‘L’ column, the problem was solved by finding the answers of the questions in ‘W’ column.

Data analysis

Quantitative data analysis techniques were employed in order to reveal if there were any statistically significant differences between problem-solving achievement pretest and posttest scores of the students in the EG, to whom the KWL reading strategy was applied and of the students in the CG,

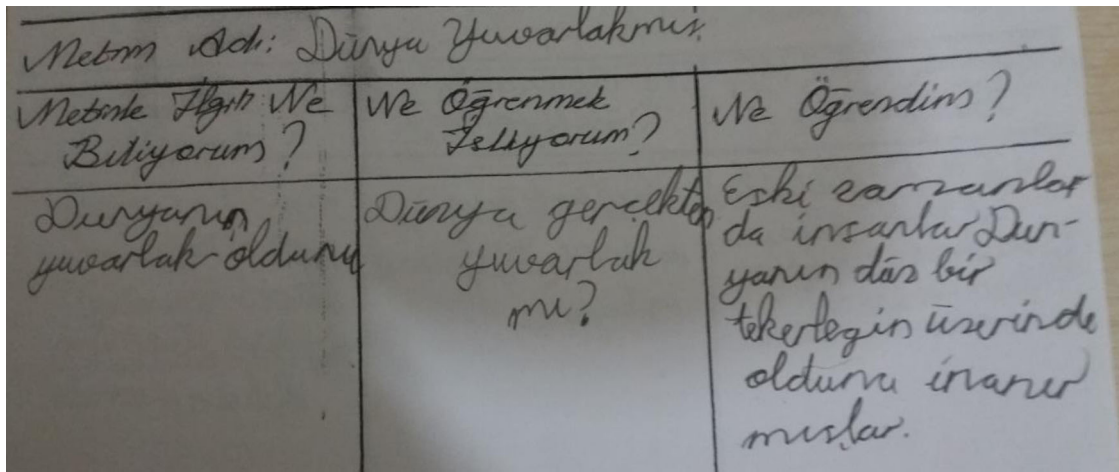


Figure 3. Study of a student in the EG about the reading text "The World is Round".

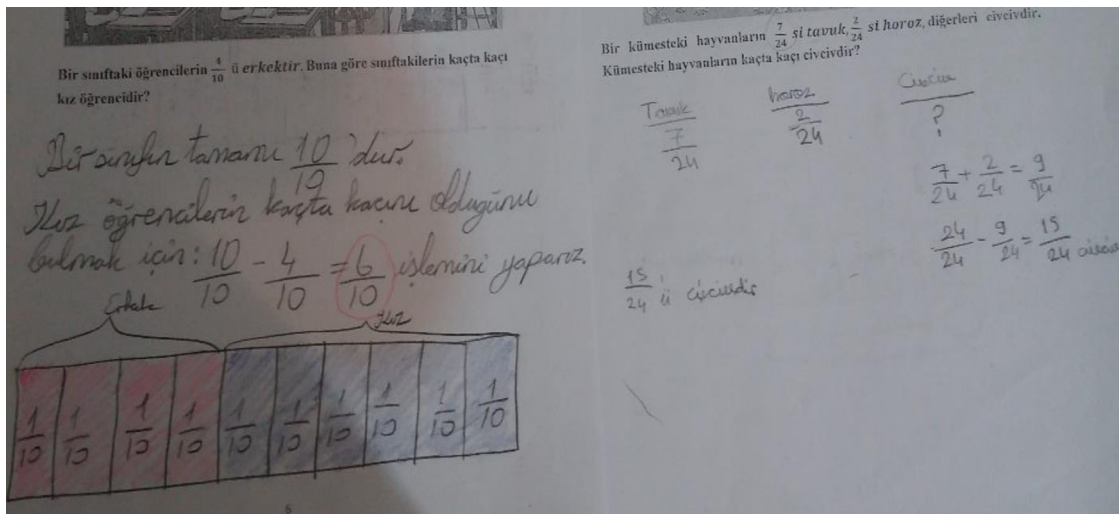


Figure 4. The mathematical problem-solving activity performed by the EG students using the KWL strategy.

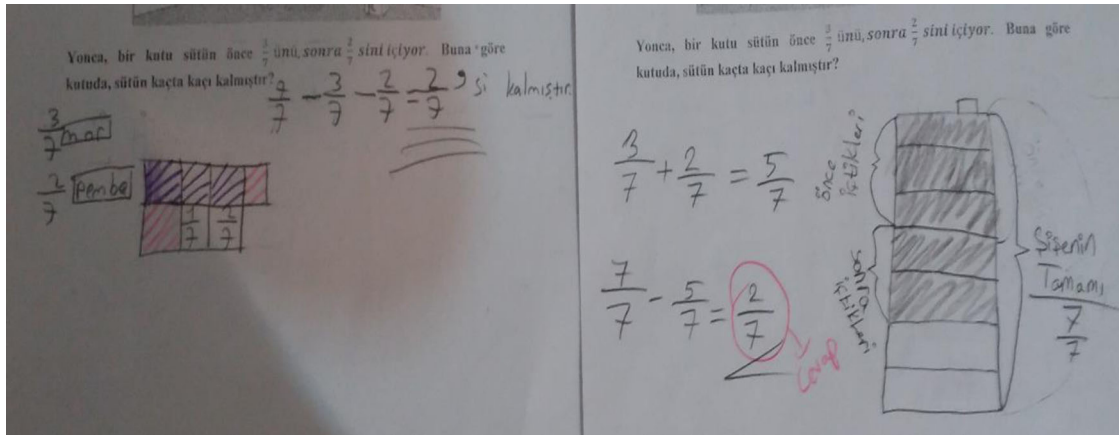


Figure 5. The mathematical problem-solving activity performed by the EG students using the KWL strategy.

to whom instruction was implemented in accordance with the current instructional program. Therefore, the questions in the problem-solving achievement test were scored using the progressive scoring scale by the two researchers, and intercoder reliability was examined according to Miles and Huberman's (1994) formula. The intercoder reliability was

found to be 89%. For the remaining 11% difference, the two researchers gathered to discuss and reached an agreement. Thus, a full correspondence (100%) was achieved by increasing the correspondence rate between the coders. Secondly, SPSS Package Program was used in order to determine whether there was a statistically significant difference

Table 3. The KWL chart related to the experimental group student's solution found by using the KWL strategy.

The KWL Chart		
K-What I Know?	W-What I Want?	L-What I Learned?
-I know what an integral is. - I know what a fraction is. - I know what unit fraction is. - I know that a fraction is each of equal units of an integral. - I know how to perform four basic operations in fractions.	-How can I express the problem in my own words? - Is there any redundant information in the problem? If yes, what? - Can I solve the problem with the information given? - How many equal parts make up the whole class? - What is the ratio of the boys in the class? - What is the ratio of the girls in the class? - Can I draw a figure illustrating all of these? - How can I solve the problem by making use of the figure? - How can I find the result of the problem? - How can I check whether I solved the problem accurately or not? - Can I write a new problem similar to this one?	-I can express the problem in my own words. "If the ratio of boys in a class is $\frac{4}{10}$, what is the ratio of girls?" - The number of students was divided into 10 equal parts. - The boys are equal to $\frac{4}{10}$ of the class. - I can draw a figure to find the ratio of the girls in the class. - If I regard the number of all students in the class as an integral, I can write the integral as $\frac{10}{10}$. - If $\frac{4}{10}$ of the class are boys, I can do $\frac{10}{10} - \frac{4}{10} = \frac{6}{10}$ to find the ratio of girls. - The result is correct because the integral was divided into 10 parts, and the boys made up 4 of 10 parts, and the girls made up 6 of 10 parts. - I can write a new problem similar to this one. I can change the numbers.

between the scores of the EG and CG students. Non-parametric techniques were employed for data analysis since the number of subjects in the groups was small. Mann-Whitney U test checks if the scores obtained from two independent samples differ significantly. This test controls if two independent groups have similar distributions in the population in terms of the concerned variable. U test requires the dependent variable to be in ordinal scale, and observations need to be independent of each other. Wilcoxon's signed ranks test is used to test significance of the difference between the scores belonging to two dependent measuring sets. This test is used as an alternative for t-test in two paired groups when variables do not show normal distribution (Büyüköztürk, 2013). The Wilcoxon's Signed Rank Test for the analyses between the dependent groups and The Mann-Whitney U Test for the analyses between the independent groups were implemented. This method was preferred since the data obtained from the research were not distributed normally, and the study group was small.

Findings

In this section of the research, the findings and interpretations were included. Firstly, the results of the test of equivalency, which was conducted to determine if the groups were equal, and descriptive analysis results were presented. Then, the sub-problems of the research were given under sections.

Results of the test of equivalency and descriptive analysis

The normality test results examined to compare equivalency of the EG and CG were presented in Table 4, and t-test results were presented in Table 5.

It was understood from Table 4 that the scores received from the test of equivalency by the experimental and control groups did not show normal distribution ($p_{EG}=.116$,

Table 4. Normality test results based on TE scores of the EG and CG.

Shapiro-Wilk			
Groups	Statistic	df	Sig
CG	.883	17	.036
EG	.917	18	.116

$p_{CG}=.036$). Thus, results of the t-test, which was used for independent samples were given in Table 5.

Independent samples t-test result regarding the scores received by the EG and CG students was $p=.655$. As the calculated p value was greater than .05, it was concluded that there were not any significant differences between the EG and CG before the implementation [$t(33) = .451$, $p=.655 > .05$]. Thus, it can be claimed that the EG and CG were equivalent before the implementation of the KWL reading strategy.

In Table 6, pretest and posttest mean scores and standard deviations of the control and experimental groups regarding the problem-solving steps and overall problem-solving achievement. It was seen that the posttest mean scores of the control group in understanding the problem ($\bar{X} = 10,41$), devising a plan ($\bar{X} = 9,59$), carrying out the plan ($\bar{X} = 12,29$), looking back ($\bar{X} = 1,35$) and overall problem-solving ($\bar{X} = 40,00$) were higher than their pretest mean scores in understanding the problem ($\bar{X} = 7,76$), devising a plan ($\bar{X} = 3,94$), carrying out the plan ($\bar{X} = 6,71$), looking back ($\bar{X} = ,88$) and overall problem-solving ($\bar{X} = 26,12$) respectively, yet their posttest mean scores in suggesting a problem ($\bar{X} = 4,71$) were lower than their pretest mean scores in the same step ($\bar{X} = 5,47$). On the other hand, the posttest mean scores of the experimental group in understanding the problem ($\bar{X} = 17,78$), devising a plan ($\bar{X} = 14,06$), carrying out the plan ($\bar{X} = 12,72$), looking back ($\bar{X} = 1,17$), suggesting a problem ($\bar{X} = 8,28$) and overall problem-solving ($\bar{X} = 55,72$) were higher than their pretest mean scores in understanding the problem ($\bar{X} = 7,22$), devising a plan ($\bar{X} = 3,22$), carrying out the plan ($\bar{X} = 5,39$), looking back ($\bar{X} = ,83$), suggesting a

Table 5. T-test results based on TE scores of the EG and CG.

Groups	N	$\bar{X}$	ss	t	p
CG	17	16.35	2.370	.451	.655
EG	18	16.67	1.715		

Table 6. Descriptive analysis results of the problem-solving steps.

Steps	Tests	Control G (N = 17)		Experimental G (N = 18)	
		$\bar{X}$	sd	$\bar{X}$	sd
Understanding the Problem	Pretest	7,76	3,011	7,22	3,209
	Posttest	10,41	4,611	17,78	2,756
Devising a Plan	Pretest	3,94	2,989	3,22	1,987
	Posttest	9,59	4,611	14,06	2,838
Carrying out the Plan	Pretest	6,71	3,820	5,39	3,127
	Posttest	12,29	3,721	12,72	2,947
Looking Back	Pretest	,88	1,764	,83	1,724
	Posttest	1,35	2,120	1,17	1,948
Suggesting a Problem	Pretest	5,47	6,848	2,33	1,749
	Posttest	4,71	6,440	8,28	5,039
Problem-Solving Achievement (Overall)	Pretest	26,12	15,252	22,22	7,674
	Posttest	40,00	16,244	55,72	11,766

problem ($\bar{X} = 2,33$) and overall problem-solving ($\bar{X} = 22,22$) respectively.

Findings and interpretations related to the first sub-problem

The Mann-Whitney U Test was employed to determine if there were any statistically significant differences between problem-solving achievement pretest scores of the EG and CG students, and the related finding was given in Table 7.

In Table 7, the Mann-Whitney U Test results according to the pretest scores received from the Progressive Scoring Scale (PSS) by the EG and CG students were shown for each step. Accordingly, no significant differences were found between two groups as a result of U test regarding the pretest scores received from the steps of PSS, which were understanding the problem ($U = 137.500$, $p > .05$), devising a plan ($U = 136.500$, $p > .05$), carrying out the plan ($U = 144.500$, $p > .05$), looking back ($U = 99.000$, $p > .05$) and suggesting a problem ($U = 144.500$, $p > .05$). Table 7 indicated that there were no significant differences between both groups regarding their pretest sum scores in all steps of problem-solving ($U = 138.500$, $p > .05$).

Findings and interpretations related to the second sub-problem

The Mann-Whitney U Test was employed to determine if there were any statistically significant differences between problem-solving achievement posttest scores of the EG and CG students, and the related finding was given in Table 8.

In Table 8, the Mann-Whitney U Test results according to the posttest scores received from the Progressive Scoring Scale (PSS) by the EG and CG students were shown for each step. Accordingly, a significant difference was found between two groups as a result of U test regarding the posttest scores received from the steps of PSS, which were

understanding the problem ($U = 17.500$, $p < .05$), devising a plan ($U = 50.500$, $p < .05$), looking back ($U = 64.000$, $p < .05$) and suggesting a problem ($U = 88.500$, $p < .05$). However, no significant difference was observed between the groups in the step of carrying out the plan ($U = 101.000$, $p > .05$). On the other hand, a significant difference was found between the two groups regarding the sum of the posttest scores received from all steps of problem-solving achievement ($U = 69.500$, $p < .05$). When mean ranks were considered, the students in the EG were found to be more successful in each of the steps of understanding the problem, devising a plan, looking back and suggesting a problem compared to the ones in the CG. In addition, mean ranks indicated that problem-solving achievement of the students in the EG was higher than the ones in the CG according to the analyses of overall sum scores. This finding revealed that instruction with the KWL reading strategy was effective in the steps of understanding the problem, devising a plan, looking back and suggesting a problem and problem-solving achievement.

Findings and interpretations related to the third sub-problem

The third sub-problem “Are there any statistically significant differences between pretest and posttest problem-solving achievement scores of the EG students on whom the KWL strategy was employed?” was tested with the Wilcoxon’s Signed Rank Test. Accordingly, the Wilcoxon’s Signed Rank Test results regarding whether there was a statistically significant difference between each step of problem-solving and the pretest and posttest scores received from the PSS before and after the implementation by the EG students were presented in Table 9.

It was seen in Table 9 that there was a significant difference between the pretest and posttest scores received from the steps of understanding the problem ($z = 3.728$, $p < .05$), devising a plan ($z = 3.724$, $p < .05$), carrying out the plan ($z = 3.584$, $p < .05$), looking back ($z = 3.628$, $p < .05$) and suggesting a problem ($z = 3.523$, $p < .05$) and problem-solving achievement ($z = 3.724$, $p < .05$) by the EG. Based on this result, it was understood that instruction with the KWL strategy developed problem-solving achievement of the EG students. According to the results of the analysis in Table 9, the KWL reading strategy was effective in fostering the EG students’ achievement in each step and in problem-solving.

Findings and interpretations related to the fourth sub-problem

The fourth sub-problem “Are there any statistically significant differences between pretest and posttest problem-solving achievement scores of the CG students on whom current instructional program was employed?” was tested with the Wilcoxon’s Signed Rank Test. Accordingly, the Wilcoxon’s Signed Rank Test results regarding whether there was a statistically significant difference between each step of problem-solving and the pretest and posttest scores

Table 7. The Mann-Whitney U pretest analysis results regarding problem-solving achievement of the EG and CG.

Problem-Solving Steps	Group	N	$\bar{X}$	sd	Mean Ranks	Rank Sum	U	z	p
Understanding the Problem	CG	17	7.76	3.011	18.91	321.50	137.500	-.516	.606
	EG	18	7.22	3.209	17.14	308.50			
Devising a Plan	CG	17	3.94	2.989	18.97	322.50	136.500	-.553	.580
	EG	18	3.22	1.987	17.08	307.50			
Carrying out the Plan	CG	17	8.06	4.023	17.53	315.50	144.500	-.283	.777
	EG	18	7.67	4.243	18.50	314.50			
Looking Back	CG	17	.88	1.764	14.82	252.00	99.000	-1.966	.049
	EG	18	1.78	1.517	21.00	378.00			
Suggesting a Problem	CG	17	5.47	6.848	18.50	314.50	144.500	-.285	.776
	EG	18	2.33	1.749	17.53	315.50			
Problem-Solving Achievement	CG	17	26.12	15.252	18.85	320.50	138.500	-.479	.632
	EG	18	22.22	7.674	17.19	309.50			

Table 8. The Mann-Whitney u posttest analysis results regarding problem-solving achievement of the EG and CG.

Problem-Solving Steps	Group	N	$\bar{X}$	sd	Mean Rank	Rank Sum	U	z	p
Understanding the Problem	CG	17	10.41	4.611	10.03	170.50	17.500	-4.493	.000
	EG	18	17.78	2.756	25.53	459.50			
Devising a Plan	CG	17	9.59	4.611	11.97	203.50	50.500	-3.401	.001
	EG	18	14.06	2.838	23.69	426.50			
Carrying out the Plan	CG	17	13.94	4.451	21.06	358.00	101.000	-1.722	.085
	EG	18	11.33	3.956	15.11	272.00			
Looking Back	CG	17	1.35	2.120	12.76	217.00	64.000	-3.101	.002
	EG	18	4.28	2.492	22.94	413.00			
Suggesting a Problem	CG	17	4.71	6.440	14.21	241.50	88.500	-2.155	.032
	EG	18	8.28	5.039	21.58	388.50			
Problem-Solving Achievement	CG	17	40.00	16.244	13.09	222.50	69.500	-2.758	.006
	EG	18	55.72	11.766	22.64	407.50			

Table 9. The Wilcoxon's signed rank test results regarding problem-solving achievement pretest and posttest scores of the EG Students.

Problem-Solving Steps	Posttest-Pretest	N	Mean Rank	Rank Sum	z	p
Understanding the Problem	Negative Rank	0	.00	.00	-3.732	.000
	Positive Rank	18	9.50	171.00		
	Equal	0				
Devising a Plan	Negative Rank	0	.00	.00	-3.737	.000
	Positive Rank	18	9.50	171.00		
	Equal	0				
Carrying out the Plan	Negative Rank	3	5.67	17.00	-2.447	.014
	Positive Rank	12	8.58	103.00		
	Equal	3				
Looking Back	Negative Rank	0	.00	.00	-3.327	.001
	Positive Rank	14	7.50	105.00		
	Equal	4				
Suggesting a Problem	Negative Rank	0	.00	.00	-3.523	.000
	Positive Rank	16	8.50	136.00		
	Equal	2				
Problem-Solving Achievement	Negative Rank	0	.00	.00	-3.724	.000
	Positive Rank	18	9.50	171.00		
	Equal	0				

*Based on negative ranks.

received from the PSS before and after the implementation by the CG students were presented in Table 10.

Table 10 indicated that there was a significant difference between the pretest and posttest scores received from the steps of devising a plan ($z = 3.305$, $p < .05$), carrying out the plan ($z = 3.538$, $p < .05$) and problem-solving achievement ($z = 3.724$, $p < .05$) by the CG. However, there were not any significant differences between their pretest and posttest scores received from the steps of understanding the problem ($z = 1.866$, $p > .05$), looking back ($z = 3.628$, $p < .05$) and suggesting a problem ($z = 3.523$, $p < .05$). Based on this result, it can be claimed that the implementation in accordance with the current instructional program helped the students be successful in problem-solving. Table 10 showed that there

was a significant difference between the pretest and posttest scores of the CG students as a result of the analyses implemented by converting the scores obtained from each step of the problem-solving to the total score. On the other hand, while the students were successful in solving mathematical problems, they were not able to show the same success for the steps of understanding the problem, looking back and suggesting a problem. It can be said that the basic understanding of obtaining this finding was that problem-solving achievement was considered as result-oriented. It was understood from Table 10 that there were not any significant differences between the test scores of the CG students obtained before and after the implementation in the most important steps of problem-solving that are understanding the

Table 10. The Wilcoxon's signed rank test results regarding problem-solving achievement pretest and posttest scores of the CG Students.

Problem-Solving Steps	Posttest-Pretest	N	Mean Rank	Rank Sum	Z	p
Understanding the Problem	Negative Rank	6	5.33	32.00	-1.866	.062
	Positive Rank	10	10.40	104.00		
	Equal	1				
Devising a Plan	Negative Rank	0	.00	.00	-3.305	.001
	Positive Rank	14	7.50	105.00		
	Equal	3				
Carrying out the Plan	Negative Rank	1	2.00	2.00	-3.538	.000
	Positive Rank	16	9.44	151.00		
	Equal	0				
Looking Back	Negative Rank	2	4.50	9.00	-.904	.366
	Positive Rank	5	3.80	19.00		
	Equal	10				
Suggesting a Problem	Negative Rank	7	6.00	42.00	-.803	.422
	Positive Rank	4	6.00	24.00		
	Equal	6				
Problem-Solving Achievement	Negative Rank	3	3.00	9.00	-3.198	.001
	Positive Rank	14	10.29	144.00		
	Equal	0				

*Based on negative ranks.

problem, looking back and suggesting a similar problem. On the contrary, the step of problem-solving was successfully carried out by the students in the CG.

Discussion and conclusion

In the present study, impact of the KWL reading strategy on problem-solving achievement of the primary school 4th graders was investigated. It was found in this study, in which a quasi-experimental design with pretest-posttest CG was employed, that the KWL strategy improved the students' problem-solving achievement as a result of non-parametric tests.

Each step of the problem-solving process was separately analyzed. Accordingly, there was a significant difference in favor of the EG in each step of problem-solving which are understanding the problem, devising a plan, looking back and suggesting a problem. Considering the sum of the scores obtained from each step, a statistically significant difference was found in problem-solving achievement of the two groups in favor of the EG. However, no significant difference was found between the scores of the EG and CG students in the step of carrying out the plan.

A significant difference was observed between the sum of the scores received from each step by the EG and CG students in favor of the former ones as a result of the within-groups analyses. On the other hand, no significant differences were found between the pretest and posttest scores received in the steps of understanding the problem, looking back and suggesting a problem by the CG students when each step was separately analyzed.

Understanding the problem is the most important step of problem-solving. This stems from the fact that a student who cannot understand the problem cannot choose and carry out an appropriate strategy for solving the problem and cannot interpret the results. The current research revealed how important the step of understanding the problem for solving a problem is. Within-group test results indicated that the results could not be appropriately interpreted by the CG students, to whom the instruction in accordance

with the activities in the current instructional program was administered. It cannot be claimed that the student can realize solution of a problem completely in situations in which the link between the given and the asked in a problem and the bridge between old and new information cannot be established. In this respect, it can be suggested that the KWL reading strategy was effective in helping the EG students understand the problem. The findings obtained from the EG students indicated that the KWL reading strategy was beneficial for solving a mathematical problem. If a student can devise a strategy for solving a problem and can implement this strategy upon understanding the problem by using the KWL strategy, the problem can be solved. An individual can produce various solutions for the problems with appropriate reasoning according to his/her own characteristics (Umay, 2003). Thus, readers need to synthesize, interpret and evaluate information from the text and use it if suitable (Wijaya & Puti, 2015). Finding a correct solution depends on appropriate reasoning and understanding the problem thoroughly because the students gather the concepts and operations during problem-solving and use them for solving the problem (Bernardo, 1999). Therefore, it can be stated that the strategy applied to the EG students contributed positively to the students' problem-solving achievements considering that the steps trigger each other. Several studies whose results are similar to the result obtained in the step of understanding the problem in the current study (Prakitipong & Nakamura, 2006; Tertemiz, 1994; Hegarty et al., 1995; Panasuk and Beyranvand, 2010, as cited by Ulu et al., 2016; Ilgın & Arslan, 2012) have been conducted. In addition, Fuentes's (1998) view that reading comprehension strategies can be used to understand the problem story due to the similarities between understanding the story and understanding the word problems has supported the result of the current study in the step of understanding the problem.

Another result of this study was that there were not any significant differences between the scores of the two groups received in the step of carrying out the plan - that is solving the problem. The reason of this situation may be that the

students did not know the ways of problem-solving. However, the results of the Wilcoxon's signed ranks test analysis which was conducted for the dependent groups revealed that there was a significant difference in the step of solving the problem. In other words, the students in both groups reached the result by solving the problems with the mathematical activities given. Another reason of this case was that the students may have had usual thoughts toward learning mathematics. This can be explained as follows. Despite the improvement studies in the curriculums conducted in Turkey from time to time, it is not realistic to expect important and sudden changes as a result of these improvements and a certain amount of time is necessary (Baki, 2014). It was observed that carrying out the plan, that is the solution of the problem, was generally carried out in the CG compared to the other steps of the problem-solving process. However, no statistically significant difference was found between the two groups. This may stem from the fact that the students just tried to find the numerical result of the problem without using mental skills such as collecting the necessary data for the solution by analyzing a problem thoroughly, deciding whether the solution is logical and adapting the problem to similar situations. This is because of the fact that the students generally regard mathematics as a structure consisting of formulas and rules and needed to be memorized. The students with such a thought adopt their teacher's solution by using his/her mathematics and algorithms. A student who is not able to produce his/her own solution and own mathematics has difficulty in solving the problem as he/she cannot transfer his/her superficial knowledge to different situations. Although our schools try to teach conceptual knowledge, functional knowledge is focused more (Baki, 2014). This leads students to be able to solve the problems they already know or have encountered before. In the current study, the reason of nonexistence of a statistically significant difference between the two groups in the step of carrying out the plan may be considered in this context.

Similarly, the EG students were more successful than the CG students in the step of suggesting a problem. With the help of the KWL strategy employed in the EG, the students were able to analyze the problems thoroughly and to transfer them mostly to similar situations. The analysis of the data obtained from the step of suggesting a problem indicated that there was a significant difference in favor of the EG.

Considering the results of the current study in general, the results of the current study are similar to the results of the studies carried out by Hite (2009), Nicolas and Emata (2018), Özdemir and Sertsöz (2006), Öztürk et al. (2020), Praktikpong and Nakamura (2006), Vilenius-Tuohimaa et al. (2008).

In the current study, the KWL strategy helped the students solve the problems by recalling their knowledge familiar with the problems via a KWL chart, determining what they wanted to learn and linking this knowledge with the knowledge they wanted to learn. Nicely's (2015) statement that the KWL chart is a reflective table indicating what

students know, what they want to learn and what they learned can be shown as evidence for his sentence. In addition, with the use of the KWL strategy, students can concentrate on analytical reading (Siribunnam & Tayraukham, 2009) and actively link the old and new knowledge (Ibrahim, 2012). Students are encouraged to take over responsibility with brainstorming in the classroom and to expand their ideas and perceptions about the topic (Szabo, 2006). In the current study, it can be stated that the KWL strategy led the students to think, reason and question in the process of problem-solving because the problem-solving process includes the processes of active questioning and thinking (Schoenfeld, 1985). In the present study, specifically impact of the KWL reading strategy on the students' problem-solving achievement was investigated. It can be indicated that the results of the current study are similar to the results of the studies revealing that reading comprehension has a positive effect on mathematics achievement and mathematical problem-solving achievement (Göktaş, 2010; Lerkkanen et al., 2005; Nicely, 2015; Pape, 2004; Passolunghi & Pazzaglia, 2005; Phromphithak, 2015; Vilenius-Tuohimaa et al., 2008).

Ulu et al. (2016) concluded that instruction of reading comprehension and problem-solving strategies raised the 5th graders' problem-solving achievement. On the other hand, Ulu et al. (2016) claimed that reading comprehension strategies could explain just 7,86% of the change in problem-solving achievement, and students' readiness level were more influential in predicting problem-solving achievement than their reading comprehension strategies. Thus, it cannot be asserted that the results of the current study conform with the results of the study carried out by Ulu et al. (2016).

The current study employed quantitative methods to examine the problem-solving achievements of the primary school 4th graders, each step of which was analyzed separately, with the KWL reading strategy. A qualitative study can be carried out in order to obtain in-depth information about reasons of these results. In addition, mixed studies in which factors affecting the problem-solving skill by using different reading strategies at different grade levels are employed can be conducted.

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Appendix 1

Test of Equivalency

9-
$$\begin{array}{r} 3642 \\ + 1965 \\ \hline \end{array}$$

İşleminin sonucunun en yakın yüzlüğü yuvarlanmış hali aşağıdakilerden hangisidir?

- a) 4610 b) 4600
c) 4800 d) 5000

10- Elimizde bulunan bir bütün ekmeği 4 eşit parçaya ayırırsak aşağıdakilerden hangisi doğru olur?

- a) Parça ekmeğe bütün ekmekten daha küçük olur.
b) Parça ekmeğe birbirine eşit olmaz.
c) Bütün ekmeğe parça ekmeğelerden daha küçük olur.
d) Her bir parça ekmeğe yarım ekmeğe olur.

11-



Ahmet'e
verilen parça

Yukarıdaki pasta 4 eşit parçaya bölünmüş ve bir parçası Ahmet'e verilmiştir.

Buna göre aşağıdakilerden hangisi doğrudur?

- a) Ahmet'e verilen pasta parçası bütün pastadan daha çoktur.
b) Bütün pasta Ahmet'e verilen pasta parçasından daha azdır.
c) Ahmet'e verilen pasta parçası bütün pastadan daha azdır.
d) Pasta parçaları eşit değildir.

12- Ahmet'in boyu 120 cm, Veli'nin boyu 110 cm, Ayşe'nin boyu ise 90 cm'dir.

Buna göre üç arkadaşın boyları toplamı kaç metredir.

- a) 3 m. b) 4 m.
c) 3 m. 20 cm d) 340 cm

13- Kader'in boyu 130 cm, Mehtap'ın boyu ise Kader'in boyundan 20 cm daha uzundur.

Buna göre Kader ile Mehtap'ın boylarının uzunluğu toplamı kaç cm'dir?

- a) 150 cm b) 280 cm
c) 3 m d) 2 m. 50 cm

14- 745 430 sayısının on binler basamağının basamak değeri kaçtır?

- a) 700 000 b) 400
c) 40 000 d) 30

15- 604 312 sayısının yüzler basamağı ile onlar basamağının sayı değerleri toplamı kaçtır?

- a) 310 b) 4312
c) 11 d) 4

16- Van-Ankara karayolu uzaklığı 1273 km'dir. Van'dan yola çıkıp otobüs 474 km gittikten sonra mola vermiştir. Otobüsün moladan sonra Ankara'ya kaç km'lik yolu kalmıştır?

- a) 890 b) 854
c) 1754 d) 799

17-

$$\begin{array}{r} 56\square 46 \\ +346\triangle 1 \\ \hline 90997 \end{array}$$

Yukarıdaki işleme göre $\square + \triangle$ kaçtır?

- a) 8 b) 7
c) 6 d) 3

18- Aşağıdaki sıralamalardan hangisi doğrudur?

- a) 124 > 1 240 > 12 400
b) 987 767 > 98 876 > 99 999
c) 201 000 > 10 100 > 1 101
d) 56 765 > 56 765 > 50 656

19- Bir kutu salça 7 Türk Lirasıdır. Market bir Hafta boyunca toplam 14 tane salça sattığına göre;

toplam kaç Türk Lirası kazanmıştır?

- a) 140 TL b) 98 TL
c) 70 TL d) 14 TL

20- Bir paket peçete içerisinde toplam 130 tane peçete bulunmaktadır. Ayşe teyze marketten 12 paket peçete aldığına göre;

bu 12 paket peçetenin içinde toplam kaç tane peçete vardır?

- a) 1560 b) 1200
c) 1300 d) 1400

Not: Her soru 5 puandır.

Not: Sınav süresi 60 dakikadır.

Cevap Anahtarı

1	(A)	(B)	(C)	(D)
2	(A)	(B)	(C)	(D)
3	(A)	(B)	(C)	(D)
4	(A)	(B)	(C)	(D)
5	(A)	(B)	(C)	(D)
6	(A)	(B)	(C)	(D)
7	(A)	(B)	(C)	(D)
8	(A)	(B)	(C)	(D)
9	(A)	(B)	(C)	(D)
10	(A)	(B)	(C)	(D)
11	(A)	(B)	(C)	(D)
12	(A)	(B)	(C)	(D)
13	(A)	(B)	(C)	(D)
14	(A)	(B)	(C)	(D)
15	(A)	(B)	(C)	(D)
16	(A)	(B)	(C)	(D)
17	(A)	(B)	(C)	(D)
18	(A)	(B)	(C)	(D)
19	(A)	(B)	(C)	(D)
20	(A)	(B)	(C)	(D)

DOĞRU

YANLIŞ

BOŞ

ALDIĞI NOT

Appendix 2

Problem-Solving Achievement Test (PSAT) (Pretest-Posttest)

1. I have 20 liras of pocket money. I bought a pen with $\frac{1}{5}$ of my pocket money and a movie ticket with $\frac{2}{5}$ of it. How many liras did I spend on my pocket money?
2. My mother read $\frac{3}{8}$ of a 400-page book. How many pages are left to be read?
3. Ms. Sevgi has planted three kinds of fruit seedlings in her garden. $\frac{2}{4}$ of the seedlings are blueberries, $\frac{1}{4}$ of strawberries and the rest are raspberries. If Ms. Sevgi has planted 10 raspberries, how many fruits has she planted in total?
4. My mother baked a cake. I ate $\frac{2}{4}$ of it while my brother ate $\frac{1}{4}$ of it. What fraction of the total cake did I and my brother eat?
5. Ahmet and Mehmet did business together. While sharing the money they earned; Ahmet said to Mehmet, "I will take $\frac{1}{4}$ of the money, and you will take $\frac{1}{5}$ ". Mehmet said, "No, no, I'll take $\frac{1}{4}$ of you, and you will take $\frac{1}{5}$ of it." said. While Ahmet and Mehmet were talking in this way, their friends Hasan came. He tried to calm them by telling them "Dear friends, why are you arguing? You will both buy three pieces, so you'll get the same amount, heal the breach!" Is Hasan right? Who wants to get more money?

6. Uncle Ahmet has an open area that he wants to divide into different compartments. He reserves half of this area for three horses. $\frac{2}{3}$ of the remaining area is marked for free movement of turkeys. The remaining area is fallowed for this summer. How much of this open area is fallowed?
7. If we serve three cakes, $\frac{3}{5}$ of one cake per plate, how many plates do we need to serve the total cake?
8. Kezban sleeps on $\frac{5}{12}$ of the day and works on $\frac{2}{24}$. How many hours did Kezban sleep and how many hours did she study during one day?
9. My mother is 27 and my father is 30 years old. My age is equal to $\frac{2}{19}$ of the total age of my parents. How old do you think I am? I would be happy if you explain your answer.

The questions of the Problem-Solving Achievement Test should be answered according to the following instructions.

- a. Express the problem in your own words.
- b. What are given in this problem? What are asked?
- c. Explain how you are going to solve this problem.
- d. Solve this problem.
- e. Check the result you have found. Is it correct? If not, what mistakes have you made? Explain your answer.
- f. Write a similar problem and solve it.

Appendix 3

Progressive Scoring Scale

Problem		
A	3	<i>Understanding the Problem</i>
	2	Understanding the problem entirely
	1	Understanding some points of the problem
	0	Being not able to understand the problem
	0	Not showing any efforts to understand the problem
B	3	<i>Devising a Plan (Choosing a Strategy)</i>
	2	Choosing a strategy that can lead to an appropriate solution
	1	Choosing only one part of strategy that can help solution
	0	Choosing an inappropriate strategy
	0	Not choosing any strategies
C	3	<i>Carrying Out the Plan</i>
	2	Finding suitable and correct solution
	1	Finding a solution which is partly correct
	0	Finding a solution which is inappropriate and incorrect
	0	Not being able to find any solutions
D	3	<i>Looking Back</i>
	2	Solving the problem and newly created problem based on it
	1	Verifying the results reasonably
	0	Verifying the results partially
	0	Not knowing how to verify the results
E	3	<i>Suggesting a Problem</i>
	2	The problem created is reasonable and solvable
	1	A new problem has been created by changing values of the problem
	0	The problem created is unreasonable and unsolvable
	0	The same or new problem has not been suggested